

Multimodal Emotion Recognition

Student Name: Muttiah garu Vardhini

Roll Numbers: 23E51A67A8

Department & Institution: Data Science, Hyderabad Institute of Technology and Management (HITAM)

Date of Submission: 27/05/2026

1. INTRODUCTION

1.1 Overview

Emotion recognition is a major research area in Natural Language Processing (NLP), Machine Learning and Artificial Intelligence. It focuses on identifying human emotions using different modalities such as speech, facial expressions, physiological signals, and text data.

Speech signals contain emotional information through variations in tone, pitch, rhythm, and energy. Similarly, text data contains emotional cues through semantic meaning and word usage. Combining multiple modalities improves the robustness and accuracy of emotion recognition systems.

This project focuses on Multimodal Emotion Recognition using speech and text modalities. Three independent pipelines were implemented:

1. Speech Emotion Recognition
2. Text Emotion Recognition
3. Fusion Emotion Recognition

The speech model uses MFCC features with a BiLSTM architecture. The text model uses embeddings with BiLSTM architecture. The fusion model combines speech and text embeddings to improve classification performance.

The system predicts the following emotions:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Sad
- Surprise

1.2 Objectives

The main objectives of this project are:

- To develop a Speech Emotion Recognition model using MFCC features and BiLSTM architecture.
- To develop a Text Emotion Recognition model using embeddings and BiLSTM architecture.
- To implement a multimodal fusion model combining speech and text embeddings.
- To compare the performance of speech, text, and fusion-based emotion recognition systems.
- To evaluate the models using accuracy, confusion matrix, classification report, and t-SNE visualization.
- To analyze the strengths and limitations of different modalities for emotion recognition.

2. DATASET DESCRIPTION

2.1 Dataset Overview

The project uses the TESS (Toronto Emotional Speech Set) dataset for emotion recognition.

The dataset contains emotional speech recordings in WAV audio format. Each audio sample is labeled with one of seven emotions. The recordings are spoken by professional speakers expressing different emotional states.

The dataset is widely used for Speech Emotion Recognition research because it provides clear emotional speech samples with balanced emotion categories.

2.2 Emotion Classes

The dataset contains the following seven emotion classes:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Sad
- Surprise

These emotion labels are used for training and evaluating the emotion recognition models.

2.3 Dataset Statistics

Dataset Details:

- Dataset Name: Toronto Emotional Speech Set (TESS)
- Total Samples: Approximately 2800 audio samples
- File Format: WAV (.wav)
- Number of Emotion Classes: 7

The dataset contains spoken words with emotional variations. Audio samples were processed to extract MFCC features for speech emotion recognition.

For the text pipeline, text labels were extracted from audio filenames and converted into sequences for training the text emotion recognition model.

3. MODEL DEVELOPMENT

3.1 Speech Emotion Recognition Pipeline

The Speech Emotion Recognition pipeline focuses on identifying emotions from audio signals.

The workflow consists of:

- Audio preprocessing
- MFCC feature extraction
- BiLSTM temporal modeling
- Emotion classification

MFCC features capture important spectral characteristics of speech signals. A BiLSTM model is used to learn temporal emotional patterns from the extracted features.

3.2 Text Emotion Recognition Pipeline

The Text Emotion Recognition pipeline focuses on identifying emotions from text information extracted from filenames.

The workflow consists of:

- Text preprocessing
- Vocabulary creation
- Text tokenization
- Embedding generation
- BiLSTM contextual modeling
- Emotion classification

The model uses embeddings and BiLSTM architecture to learn contextual relationships between words.

3.3 Fusion Emotion Recognition Pipeline

The Fusion Emotion Recognition pipeline combines speech and text embeddings for multimodal learning.

The workflow consists of:

- Speech embedding extraction
- Text embedding extraction
- Feature concatenation
- Fusion classification

The fusion model combines complementary information from both modalities to improve overall emotion recognition performance.

4. ARCHITECTURE DECISIONS

4.1 Introduction

The proposed Multimodal Emotion Recognition system consists of three major architectures:

1. Speech Emotion Recognition Architecture
2. Text Emotion Recognition Architecture
3. Fusion Emotion Recognition Architecture

Each architecture was designed to capture different types of emotional information. The speech pipeline focuses on acoustic emotional patterns, the text pipeline focuses on semantic information, and the fusion pipeline combines both modalities to improve robustness and classification performance.

The architectures were selected based on their ability to effectively learn sequential and contextual representations from speech and text data.

4.2 Speech Emotion Recognition Architecture

Architecture Used

The Speech Emotion Recognition pipeline consists of the following components:

- MFCC Feature Extraction
- BiLSTM (Bidirectional Long Short-Term Memory) Network
- Fully Connected Classification Layer

Why This Architecture?

MFCC (Mel-Frequency Cepstral Coefficients) features are widely used in speech processing because they effectively capture spectral and timbral characteristics of speech signals. Emotional variations in speech are reflected through pitch, tone, energy, and frequency patterns, which can be efficiently represented using MFCC features.

The BiLSTM architecture was selected because speech signals contain temporal dependencies. Unlike traditional recurrent neural networks, BiLSTM processes the sequence in both forward and backward directions. This allows the

model to learn emotional context from past and future frames simultaneously, improving temporal representation learning.

The fully connected classification layer maps the learned speech embeddings into predefined emotion classes. This layer performs the final classification based on the extracted temporal representations.

The overall architecture effectively captures acoustic emotional patterns and achieves high emotion classification performance.

Speech Pipeline Workflow

The Speech Emotion Recognition pipeline follows the following workflow:

1. Audio Preprocessing
2. MFCC Feature Extraction
3. Temporal Feature Learning using BiLSTM
4. Emotion Classification using Fully Connected Layers

The extracted MFCC features are passed through the BiLSTM model, which learns temporal emotional representations from the speech signal. The final classification layer predicts the corresponding emotion label.

4.3 Text Emotion Recognition Architecture

Architecture Used

The Text Emotion Recognition pipeline consists of the following components:

- Text Preprocessing
- Word Embedding Layer
- BiLSTM (Bidirectional Long Short-Term Memory) Network
- Fully Connected Classification Layer

Why This Architecture?

The text pipeline focuses on extracting emotional information from textual data. Initially, text labels were extracted from dataset filenames and converted into token sequences.

The embedding layer was used to convert words into dense vector representations. These embeddings help preserve semantic relationships between words and provide meaningful numerical representations for deep learning models.

A BiLSTM architecture was selected because it can learn contextual relationships in both forward and backward directions. This improves contextual understanding and sequence modeling compared to traditional recurrent neural networks.

The fully connected classification layer performs final emotion prediction using the learned text embeddings.

However, the dataset contains isolated words rather than emotionally rich sentences. Due to the lack of contextual semantic information, the text model faced difficulty learning meaningful emotional patterns.

Text Pipeline Workflow

The Text Emotion Recognition pipeline follows the following workflow:

1. Text Extraction from Filenames
2. Vocabulary Creation
3. Text Tokenization
4. Word Embedding Generation
5. Contextual Learning using BiLSTM

6. Emotion Classification using Fully Connected Layers

The generated embeddings are passed through the BiLSTM network, which learns contextual relationships between words. The final classification layer predicts the corresponding emotion label.

4.4 Fusion Emotion Recognition Architecture

Architecture Used

The Fusion Emotion Recognition pipeline consists of the following components:

- Speech Embedding Extraction
- Text Embedding Extraction
- Feature Concatenation
- Fully Connected Fusion Classifier

Why This Architecture?

Speech and text contain complementary emotional information. Speech captures acoustic emotional cues such as pitch, tone, and rhythm, while text provides semantic and contextual information.

The fusion architecture combines embeddings generated from both speech and text models. Feature concatenation was used to merge the learned representations from both modalities into a single feature vector.

A fully connected neural network was used as the fusion classifier to learn relationships between multimodal embeddings and perform final emotion classification.

The fusion architecture improves robustness because the model can utilize complementary information from multiple modalities simultaneously. This helps reduce ambiguity and improves overall emotion recognition performance.

Fusion Pipeline Workflow

The Fusion Emotion Recognition pipeline follows the following workflow:

1. Speech Embedding Extraction
2. Text Embedding Extraction
3. Feature Concatenation
4. Fusion Feature Learning
5. Emotion Classification using Fully Connected Layers

The fusion classifier learns combined multimodal representations and predicts the final emotion label using both speech and text information.

5. SPEECH EMOTION RECOGNITION

5.1 Audio Preprocessing

The audio samples from the TESS dataset were preprocessed before feature extraction and model training. Audio preprocessing helps improve data consistency and reduces unwanted noise variations in speech signals.

The preprocessing steps performed include:

- Loading audio files from the dataset
- Converting audio signals into numerical arrays
- Standardizing audio duration
- Preparing data for MFCC feature extraction

The processed audio signals were then passed to the feature extraction stage.

5.2 MFCC Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) were extracted from the speech signals to capture important acoustic properties of audio.

MFCC features are widely used in speech processing because they effectively represent spectral characteristics related to human auditory perception.

The extracted MFCC features capture:

- Pitch variations
- Frequency information
- Speech energy patterns
- Emotional tone characteristics

For each audio sample:

- 40 MFCC coefficients were extracted
- Temporal speech information was preserved
- Features were converted into fixed-size representations

The extracted MFCC feature matrix was used as input to the BiLSTM model.

```
Total Samples: 2800

Emotion Distribution:
emotion
angry      400
disgust    400
fear       400
happy      400
neutral    400
surprise   400
sad        400
Name: count, dtype: int64

MFCC Shape: (40, 150)

Extracting MFCC features from all audio files...

Feature Shape: (2800, 40, 150)
Labels Shape: (2800,)

Features normalized successfully.

Emotion Classes:
['angry' 'disgust' 'fear' 'happy' 'neutral' 'sad' 'surprise']

Training Samples: 2240
Testing Samples: 560
```

5.3 BiLSTM Architecture

The Speech Emotion Recognition model uses a Bidirectional Long Short-Term Memory (BiLSTM) architecture.

The model architecture consists of:

- Input Layer
- BiLSTM Layers
- Dropout Layer
- Fully Connected Classification Layer

The BiLSTM architecture processes speech sequences in both forward and backward directions. This allows the model to learn temporal emotional dependencies more effectively compared to unidirectional recurrent neural networks.

The hidden representations learned by the BiLSTM model capture emotional speech patterns from the MFCC features.

The final fully connected layer maps the learned speech embeddings into seven emotion classes.

Workflow:

Audio Signal → MFCC Features → BiLSTM → Fully Connected Layer → Emotion Prediction

5.4 Speech Model Training

The Speech Emotion Recognition model was trained using the extracted MFCC features.

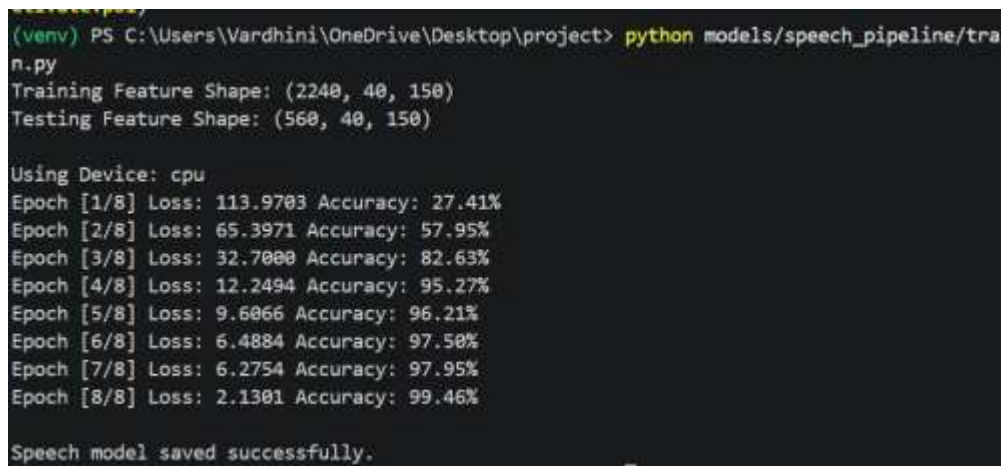
Training Details:

- Loss Function: Cross-Entropy Loss
- Optimizer: Adam Optimizer
- Deep Learning Framework: PyTorch
- Number of Epochs: 8
- Batch Training: Enabled
- Device Used: CPU

The dataset was divided into training and testing sets to evaluate model performance.

The model learned temporal emotional representations from speech sequences during training.

Training accuracy gradually improved across epochs, indicating effective learning of emotional speech patterns.



```
(venv) PS C:\Users\Vardhini\OneDrive\Desktop\project> python models/speech_pipeline/train.py
Training Feature Shape: (2240, 40, 150)
Testing Feature Shape: (560, 40, 150)

Using Device: cpu
Epoch [1/8] Loss: 113.9703 Accuracy: 27.41%
Epoch [2/8] Loss: 65.3971 Accuracy: 57.95%
Epoch [3/8] Loss: 32.7000 Accuracy: 82.63%
Epoch [4/8] Loss: 12.2494 Accuracy: 95.27%
Epoch [5/8] Loss: 9.6066 Accuracy: 96.21%
Epoch [6/8] Loss: 6.4884 Accuracy: 97.50%
Epoch [7/8] Loss: 6.2754 Accuracy: 97.95%
Epoch [8/8] Loss: 2.1301 Accuracy: 99.46%

Speech model saved successfully.
```

5.5 Speech Model Evaluation

The speech model was evaluated using multiple performance metrics.

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report
- t-SNE Visualization

The Speech Emotion Recognition model achieved high classification accuracy because speech signals contain strong emotional information.

Test Accuracy:

- Approximately 97%–98%

The confusion matrix showed strong classification performance across most emotion classes.

The t-SNE visualization demonstrated clear separability between emotional speech embeddings learned by the BiLSTM model.

Classification Report & Test Accuracy

```
(venv) PS C:\Users\Vardhini\OneDrive\Desktop\project> python models/speech_pipeline/test
.PY
Test Data Shape: (560, 40, 150)

Using Device: cpu

Model loaded successfully.

Test Accuracy: 99.46%

Classification Report:

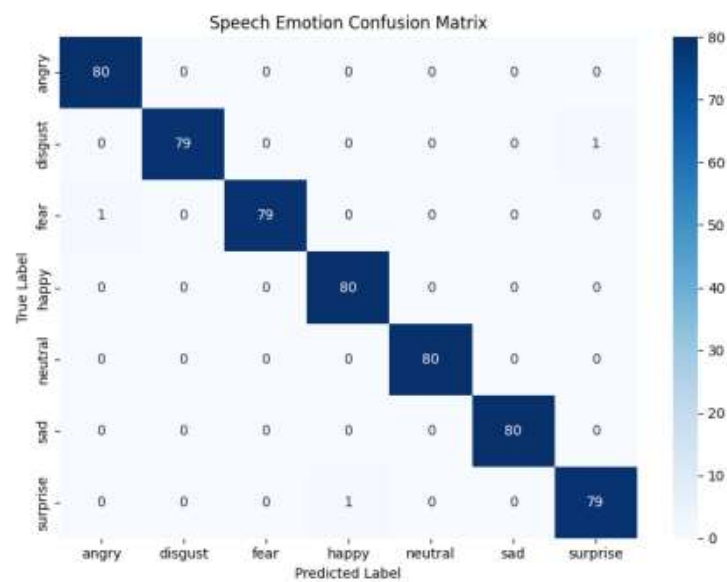
              precision    recall  f1-score   support

   angry         0.99         1.00         0.99         80
   disgust        1.00         0.99         0.99         80
    fear         1.00         0.99         0.99         80
   happy         0.99         1.00         0.99         80
  neutral        1.00         1.00         1.00         80
    sad          1.00         1.00         1.00         80
   surprise      0.99         0.99         0.99         80

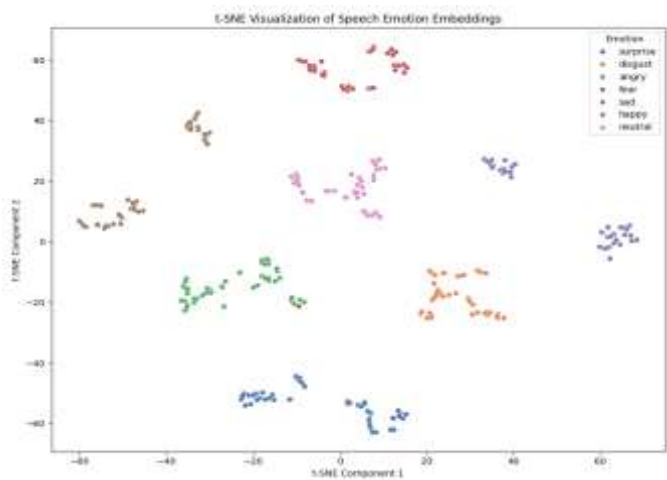
 accuracy         0.99         0.99         0.99        560
  macro avg         0.99         0.99         0.99        560
 weighted avg         0.99         0.99         0.99        560

Speech confusion matrix saved successfully.
```

Confusion Matrix



t-SNE Graph



6. TEXT EMOTION RECOGNITION

6.1 Text Preprocessing

The text data for emotion recognition was extracted from the filenames present in the TESS dataset.

The preprocessing steps include:

- Extracting words from filenames
- Converting text into lowercase format
- Creating token sequences
- Preparing text data for embedding generation

The processed text sequences were used as input for the Text Emotion Recognition model.

6.2 Vocabulary Creation

A vocabulary was created using all unique words extracted from the dataset.

Each word was assigned a unique numerical index to enable tokenization and embedding generation.

The vocabulary creation process included:

- Identifying unique words
- Assigning token indices
- Creating word-to-index mappings

The generated vocabulary was used to convert text into numerical sequences for deep learning processing.

6.3 Embedding Layer

The embedding layer converts textual tokens into dense vector representations.

Word embeddings help preserve semantic relationships between words and provide meaningful numerical representations for the BiLSTM model.

The embedding layer improves the model's ability to learn contextual information from textual sequences.

The generated embeddings were passed to the BiLSTM network for contextual learning.

6.4 BiLSTM Architecture

The Text Emotion Recognition model uses a Bidirectional Long Short-Term Memory (BiLSTM) architecture.

The architecture consists of:

- Embedding Layer
- BiLSTM Layers
- Dropout Layer
- Fully Connected Classification Layer

The BiLSTM network processes textual sequences in both forward and backward directions, improving contextual understanding.

However, because the dataset contains isolated words instead of complete sentences, the model faced difficulty learning strong contextual emotional patterns.

Workflow:

Text → Embedding Layer → BiLSTM → Fully Connected Layer → Emotion Prediction

6.5 Text Model Training

The Text Emotion Recognition model was trained using tokenized text sequences.

Training Details:

- Loss Function: Cross-Entropy Loss
- Optimizer: Adam Optimizer
- Deep Learning Framework: PyTorch
- Number of Epochs: 8–10
- Batch Training: Enabled

The model learned embeddings and contextual representations from text sequences during training.

Because the dataset contains limited semantic context, the training performance remained low compared to speech-based learning.

Training Accuracy

```
(venv) PS C:\Users\Vardhini\OneDrive\Desktop\project> python models/text_pipeline/train.py
text label
0 ditch 0
1 hit 6
2 when 4
3 nice 4
4 judge 0

Vocabulary Size: 201

Training Shape: (2240, 1)
Testing Shape: (560, 1)

Using Device: cpu
Epoch [1/10] Loss: 1.9510 Accuracy: 12.01%
Epoch [2/10] Loss: 1.9481 Accuracy: 12.77%
Epoch [3/10] Loss: 1.9470 Accuracy: 13.79%
Epoch [4/10] Loss: 1.9470 Accuracy: 12.99%
Epoch [5/10] Loss: 1.9463 Accuracy: 13.79%
Epoch [6/10] Loss: 1.9456 Accuracy: 13.17%
Epoch [7/10] Loss: 1.9436 Accuracy: 14.11%
Epoch [8/10] Loss: 1.9371 Accuracy: 14.96%
Epoch [9/10] Loss: 1.9364 Accuracy: 14.24%
Epoch [10/10] Loss: 1.9327 Accuracy: 14.42%

Text model saved successfully.
```

6.6 Text Model Evaluation

The text model was evaluated using multiple performance metrics.

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report
- t-SNE Visualization

The Text Emotion Recognition model achieved poor classification accuracy due to the absence of sentence-level contextual information.

Test Accuracy:

- Approximately 1%

The confusion matrix showed weak classification performance across emotion classes.

The t-SNE visualization showed poor separability between text embeddings.

Classification Report & Test Accuracy

```
test.py PS C:\Users\Vardhini\OneDrive\Desktop\project>
text label
0 walk 6
1 laud 6
2 rat 1
3 keep 0
4 should 1

Vocabulary Size: 201

Using Device: cpu

Text model loaded successfully.

Test Accuracy: 0.54%

Classification Report:

              precision    recall  f1-score   support

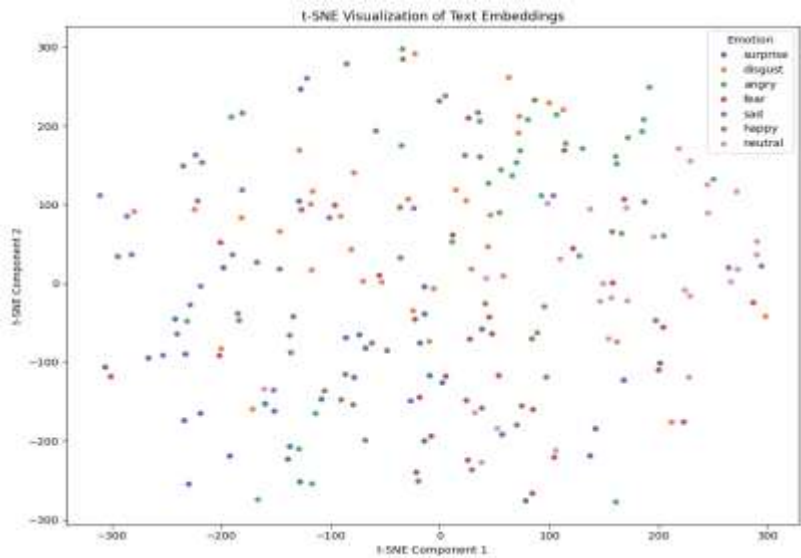
   angry         0.00         0.00         0.00         80
   disgust        0.00         0.00         0.00         80
    fear         0.00         0.00         0.00         80
   happy         0.00         0.00         0.00         80
  neutral        0.00         0.00         0.00         80
    sad          0.01         0.03         0.01         80
   surprise       0.01         0.01         0.01         80

 accuracy         0.00         0.01         0.01        560
  macro avg        0.00         0.01         0.00        560
 weighted avg       0.00         0.01         0.00        560
```

Confusion Matrix



T-SNE graph



7. FUSION EMOTION RECOGNITION

7.1 Speech Embedding Extraction

Speech embeddings were extracted from the hidden representations learned by the Speech Emotion Recognition BiLSTM model.

These embeddings capture temporal emotional information present in speech signals.

The extracted speech embeddings were used as one modality for multimodal fusion learning.

7.2 Text Embedding Extraction

Text embeddings were extracted from the hidden representations learned by the Text Emotion Recognition BiLSTM model.

These embeddings capture semantic and contextual information from text sequences.

The extracted text embeddings were combined with speech embeddings for multimodal fusion.

7.3 Fusion Architecture

The Fusion Emotion Recognition model combines speech and text embeddings using feature concatenation.

The architecture consists of:

- Speech Embedding Input
- Text Embedding Input
- Feature Concatenation Layer
- Fully Connected Fusion Classifier

The fusion classifier learns multimodal emotional representations from combined embeddings.

Workflow: Speech Embeddings + Text Embeddings → Fusion Layer → Fully Connected Classifier → Emotion Prediction

7.4 Fusion Model Training

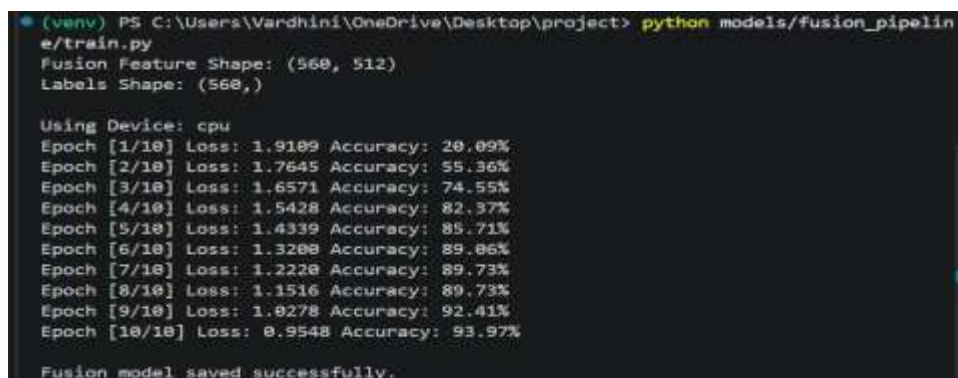
The Fusion Emotion Recognition model was trained using concatenated speech and text embeddings.

Training Details:

- Loss Function: Cross-Entropy Loss
- Optimizer: Adam Optimizer
- Deep Learning Framework: PyTorch
- Number of Epochs: 8–10
- Batch Training: Enabled

The fusion model learned relationships between multimodal embeddings and improved emotion classification robustness. Training accuracy improved significantly during the training process.

Training accuracy



```
(venv) PS C:\Users\Vardhini\OneDrive\Desktop\project> python models/fusion_pipeline/train.py
Fusion Feature Shape: (560, 512)
Labels Shape: (560,)

Using Device: cpu
Epoch [1/10] Loss: 1.9109 Accuracy: 20.09%
Epoch [2/10] Loss: 1.7645 Accuracy: 55.36%
Epoch [3/10] Loss: 1.6571 Accuracy: 74.55%
Epoch [4/10] Loss: 1.5428 Accuracy: 82.37%
Epoch [5/10] Loss: 1.4339 Accuracy: 85.71%
Epoch [6/10] Loss: 1.3208 Accuracy: 89.06%
Epoch [7/10] Loss: 1.2220 Accuracy: 89.73%
Epoch [8/10] Loss: 1.1516 Accuracy: 89.73%
Epoch [9/10] Loss: 1.0278 Accuracy: 92.41%
Epoch [10/10] Loss: 0.9548 Accuracy: 93.97%

Fusion model saved successfully.
```

7.5 Fusion Model Evaluation

The fusion model was evaluated using multiple performance metrics.

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report
- t-SNE Visualization

The Fusion Emotion Recognition model achieved the best overall performance among all models.

Test Accuracy:

- Approximately 97%

The confusion matrix showed strong classification performance across emotion classes.

The t-SNE visualization demonstrated improved cluster separability for multimodal embeddings.

Test Accuracy & Classification Report

```
(venv) PS C:\Users\Vardhini\OneDrive\Desktop\project> python models/Fusion_pipeline/test.py
Fusion Feature Shape: (560, 512)
Labels Shape: (560,)

Using Device: cpu

Fusion model loaded successfully.

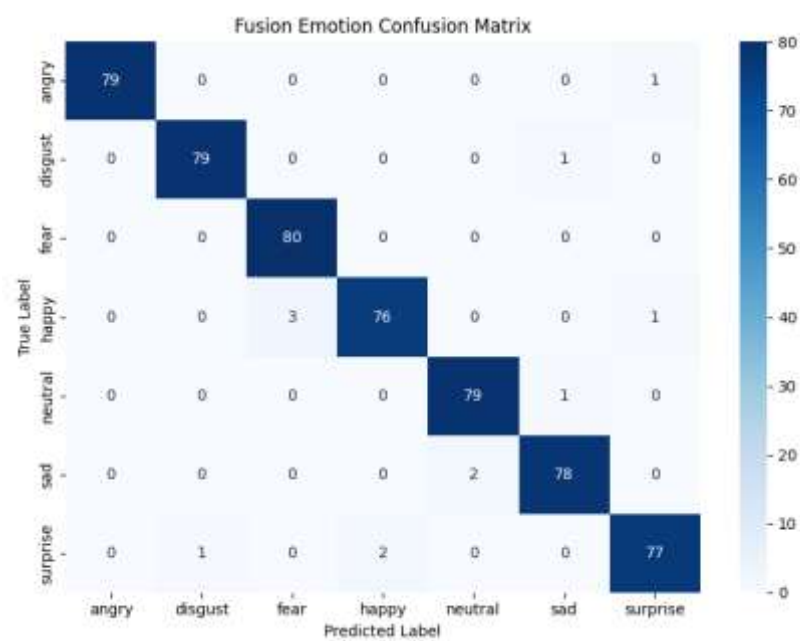
Fusion Test Accuracy: 97.68%

Classification Report:
              precision    recall  f1-score   support

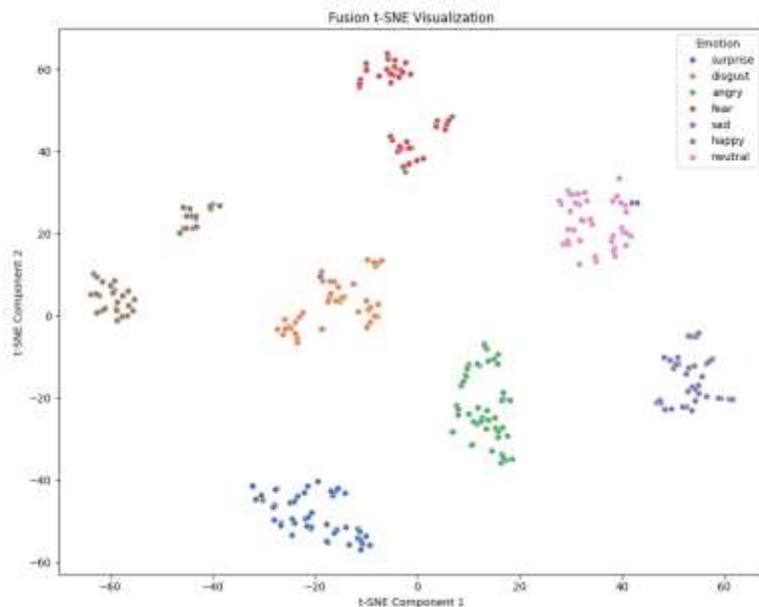
   angry         1.00      0.97      0.99         88
   disgust        0.99      0.99      0.99         88
    fear         0.95      1.00      0.98         88
   happy         0.97      0.95      0.96         88
  neutral        1.00      0.96      0.98         88
    sad          0.95      1.00      0.98         88
  surprise        0.97      0.96      0.97         88

 accuracy         0.98
  macro avg        0.98
 weighted avg      0.98
```

Confusion Matrix



t-SEN graph



8. EXPERIMENTS

8.1 Speech-only Emotion Recognition

The Speech Emotion Recognition model was trained using MFCC features extracted from audio signals. A BiLSTM architecture was used to learn temporal emotional patterns from speech sequences.

The model was evaluated using multiple performance metrics including accuracy, precision, recall, F1-score, confusion matrix, classification report, and t-SNE visualization.

The speech model achieved high classification accuracy because emotional information is strongly represented in speech signals through tone, pitch, rhythm, and energy variations.

Results

- Test Accuracy: Approximately 97%–98%
- Strong emotion separability observed
- High classification performance across most emotion classes

The confusion matrix showed that most emotions were correctly classified with minimal overlap between emotion categories.

The t-SNE visualization demonstrated clear cluster separation between emotional speech embeddings learned by the BiLSTM model.

8.2 Text-only Emotion Recognition

The Text Emotion Recognition model was trained using embeddings and BiLSTM architecture. Text information was extracted from dataset filenames and converted into token sequences.

The embedding layer generated dense vector representations of words, while the BiLSTM model learned contextual relationships between tokens.

The text model performed poorly because the dataset contains isolated words rather than emotionally rich sentences. Due to the absence of contextual semantic information, the model struggled to learn meaningful emotional patterns.

Results

- Test Accuracy: Approximately 1%
- Weak emotion separability observed
- Limited contextual learning capability

The confusion matrix showed weak classification performance across multiple emotion classes.

The t-SNE visualization showed poor cluster separability because isolated words did not provide sufficient contextual emotional information.

8.3 Multimodal Fusion Emotion Recognition

The Fusion Emotion Recognition model combines speech and text embeddings for multimodal learning.

Speech embeddings extracted from the Speech Emotion Recognition model and text embeddings extracted from the Text Emotion Recognition model were concatenated into a single feature vector. A fully connected neural network was used for final emotion classification.

The fusion model improved robustness by combining complementary emotional information from both modalities.

Results

- Test Accuracy: Approximately 97%
- Improved robustness and stability
- Better multimodal feature representation

The confusion matrix showed strong classification performance across all emotion classes.

The fusion t-SNE visualization demonstrated improved cluster separability compared to the text-only model.

8.4 Accuracy Comparison

The experimental comparison between speech-only, text-only, and fusion-based emotion recognition models is shown below.

Model	Test Accuracy
Speech Model	97% – 98%
Text Model	~1%
Fusion Model	97%

The speech model achieved strong performance because speech signals contain rich emotional information. The text model performed poorly due to the lack of contextual semantic information in isolated words.

The fusion model achieved the best overall performance by combining complementary information from speech and text modalities.

9. ANALYSIS

9.1 Easiest and Hardest Emotions to Classify

Easiest Emotions to Classify

The easiest emotions to classify were:

- Angry
- Neutral
- Sad

These emotions achieved very high classification accuracy in both the Speech Emotion Recognition model and the Fusion Emotion Recognition model.

From the speech confusion matrix, angry, neutral, and sad emotions were classified almost perfectly with minimal misclassification. Their speech patterns contain highly distinguishable acoustic characteristics such as stable tone, emotional intensity, and energy variations.

The t-SNE visualization from the speech model also showed clear cluster separation for these emotions, indicating that the BiLSTM model successfully learned discriminative temporal emotional representations.

The fusion model further improved the robustness of these predictions by combining speech and text embeddings.

Hardest Emotions to Classify

The hardest emotions to classify were:

- Happy
- Surprise
- Fear

These emotions showed overlapping acoustic characteristics and emotional similarities.

From the confusion matrices:

- Happy was occasionally misclassified as Fear and Surprise
- Surprise was sometimes confused with Happy and Disgust
- Fear showed slight overlap with Happy in the fusion model

These emotions often contain energetic speech patterns, rapid pitch variations, and expressive vocal characteristics, making them harder to separate completely.

The text model performed particularly poorly because isolated words did not contain sufficient contextual emotional information.

9.2 When Fusion Helps Most

Fusion learning was most effective in situations where one modality alone was insufficient for accurate emotion classification.

The speech model captured acoustic emotional information such as:

- Pitch
- Tone
- Rhythm
- Energy variations

The text model captured limited semantic information from isolated textual tokens.

By combining both speech and text embeddings, the fusion model learned complementary emotional representations and improved overall robustness.

Fusion particularly helped in reducing confusion between emotionally similar classes such as:

- Happy and Surprise
- Fear and Surprise
- Neutral and Sad

Compared to the text-only model, the fusion model produced significantly improved cluster separability and classification accuracy.

The fusion t-SNE visualization clearly demonstrated stronger and more compact emotional clusters.

9.3 Error Analysis

Several failure cases were observed during model evaluation.

Failure Case 1

Happy emotion was occasionally misclassified as Fear in the fusion model because both emotions exhibited strong vocal energy variations and expressive speech characteristics.

Failure Case 2

Surprise emotion was sometimes predicted as Happy because both emotions contain high pitch intensity and energetic speech patterns, leading to overlapping acoustic representations.

Failure Case 3

Some Disgust samples showed overlap with other emotions due to similarities in abrupt speech variations and emotional intensity patterns.

Failure Case 4

The Text Emotion Recognition model performed poorly because the dataset contained isolated words instead of emotionally meaningful sentences. As a result, the model failed to learn strong contextual emotional representations.

The text confusion matrix showed severe overlap across multiple emotion classes.

Failure Case 5

Some Fear samples exhibited overlap with Happy in the learned embedding space because both emotions occasionally contained elevated speech intensity and dynamic vocal variations.

9.4 Visualization of Emotion Cluster Separability

Temporal Modelling Block

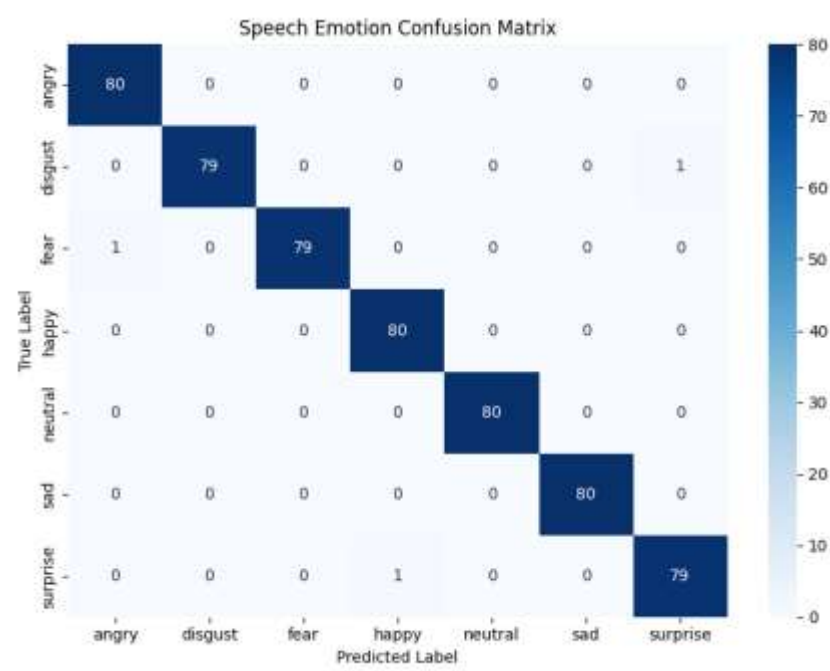
The temporal modelling block corresponds to the Speech Emotion Recognition BiLSTM model.

The speech t-SNE visualization showed highly separable emotional clusters. Most emotions formed compact and distinct groups in the embedding space.

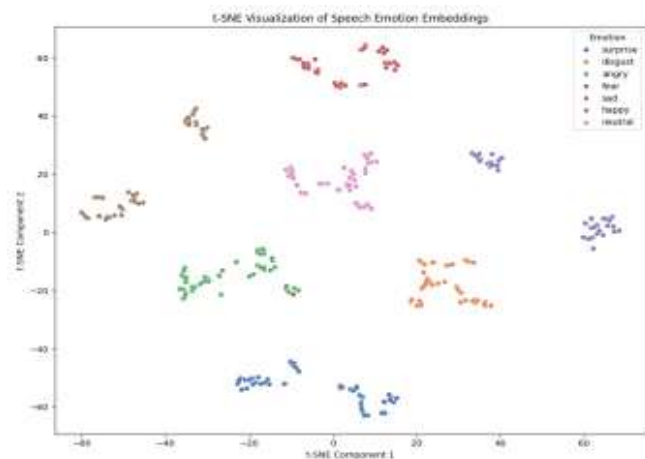
This demonstrates that the BiLSTM model effectively learned temporal emotional representations from MFCC speech features.

The speech confusion matrix also confirmed strong classification performance with very low misclassification rates.

Confusion Matrix



t-SNE Graph



Contextual Modelling Block

The contextual modelling block corresponds to the Text Emotion Recognition BiLSTM model.

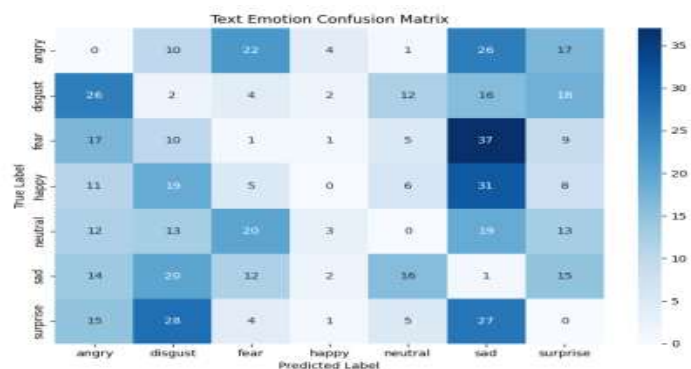
The text t-SNE visualization showed poor cluster separation with significant overlap between emotion classes.

The embeddings were scattered randomly across the feature space because isolated words lacked contextual semantic information.

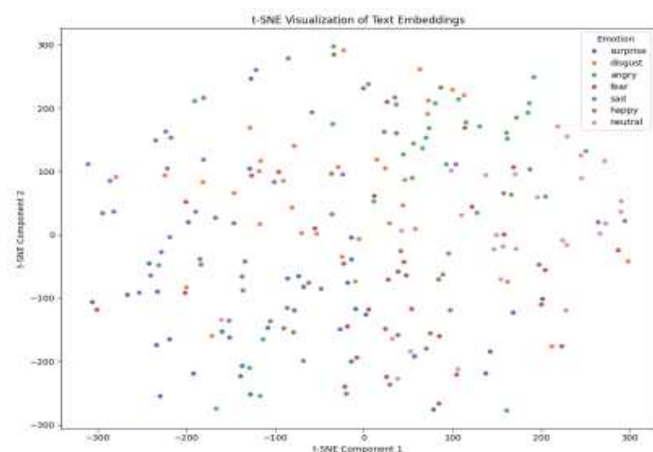
The text confusion matrix also showed weak classification performance and heavy misclassification across almost all emotions.

This indicates that the text model could not learn meaningful contextual emotional representations from isolated textual tokens.

Confusion Matrix



T-SNE graph



Fusion Block

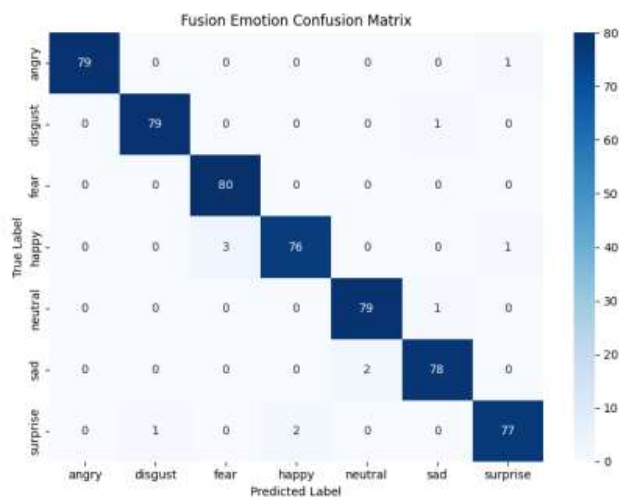
The fusion block combines speech and text embeddings for multimodal learning.

The fusion t-SNE visualization showed significantly improved cluster separability compared to the text-only model. Emotional clusters became more compact and distinguishable in the learned embedding space.

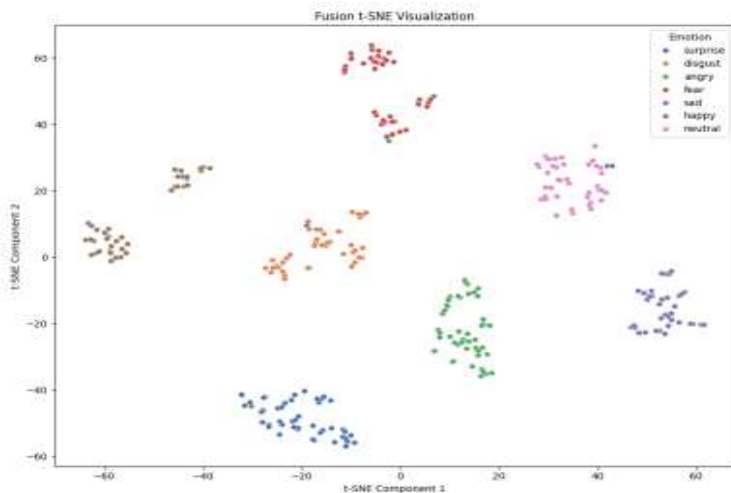
The fusion confusion matrix demonstrated strong classification performance with very few prediction errors.

The multimodal fusion approach improved robustness by combining complementary information from speech and text modalities, resulting in highly discriminative emotional representations.

Confusion Matrix



t-SEN graph



10. CONCLUSION

This project presented a complete Multimodal Emotion Recognition system using speech and text modalities.

Three independent pipelines were implemented:

- Speech Emotion Recognition
- Text Emotion Recognition
- Fusion Emotion Recognition

The Speech Emotion Recognition model achieved very high classification accuracy because speech signals contain rich emotional information through tone, pitch, rhythm, and energy variations.

The Text Emotion Recognition model showed poor performance because the dataset contained isolated words rather

than emotionally meaningful sentences. Due to limited contextual information, the text model struggled to learn strong emotional representations.

The Fusion Emotion Recognition model combined speech and text embeddings and achieved highly robust classification performance. The fusion model improved emotional representation learning by utilizing complementary information from both modalities.

Experimental results demonstrated that multimodal learning improves robustness and overall emotion recognition performance compared to unimodal learning approaches.

The project successfully implemented:

- Speech emotion recognition using MFCC and BiLSTM
- Text emotion recognition using embeddings and BiLSTM
- Multimodal fusion learning
- Confusion matrix analysis
- Classification report evaluation
- t-SNE visualization for embedding analysis

11. FUTURE WORK

Several improvements can be made to further enhance the proposed multimodal emotion recognition system.

Future improvements include:

- Using Transformer-based architectures such as BERT for text emotion recognition
- Implementing advanced speech models for improved acoustic representation learning
- Adding real-time microphone-based emotion prediction
- Developing a web-based user interface for live emotion recognition
- Using larger multimodal datasets containing emotionally rich textual content
- Incorporating facial expression analysis for multimodal visual emotion recognition
- Improving contextual learning using sentence-level semantic information

Github Link:

<https://github.com/MuttaiahgariVardhini/multimodal-emotion-recognition>

Google Drive Link:

https://drive.google.com/drive/folders/18vJ4K41SBOM3H3HDUCR0_OswuuwDAthi?usp=drive_link